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Effect of repetitive transcranial magnetic stimulation on quadriceps function in participants in the early postoperative period after anterior cruciate ligament reconstruction: a randomized controlled trial

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Abstract

Background Insufficient quadriceps activation is likely to occur in the early postoperative period after anterior cruciate ligament reconstruction (ACLR). Transcranial magnetic stimulation (TMS) has a positive effect on the activation and contractility of the quadriceps femoris. To investigate the effect of repetitive transcranial magnetic stimulation (rTMS) on quadriceps femoris function in participants during the early postoperative period following anterior cruciate ligament reconstruction (ACLR).

Methods Fifty six participants received rehabilitation education and a recommended routine training and knee flexion and extension exercise plan. In addition to this, the REAL group received 2 weeks of transcranial magnetic stimulation treatment. The resting motor threshold (RMT), root mean square (RMS) value, difference in knee extension angle between the healthy and affected sides, visual analog scale (VAS) score, and difference in the circumference of the vastus medialis oblique between the healthy and affected sides were assessed at baseline and 2 weeks after treatment.

Results Compared with those in the SHAM group, the improvements in the RMT and RMS value of the vastus medialis oblique, the difference in the knee extension angle between the healthy and affected sides, and the VAS score in the REAL group were more significant ($p < 0.001$). There was no significant between-group difference in the difference in the circumference of the vastus medialis oblique between the healthy and affected sides ($p > 0.05$).

Conclusion The use of rTMS, the goal of which is to increase the excitability of the corticospinal pathway, can promote the activation of the quadriceps femoris, prevent muscle atrophy, and accelerate the recovery of muscle strength in participants in the early postoperative period after ACLR.

Trial registration The Registration number of the Chinese Clinical Trial Registry is ChiCTR2300076522, 11/10/2023.

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Keywords Anterior cruciate ligament reconstruction, Early rehabilitation, Repetitive transcranial magnetic stimulation, Quadriceps femoris

Background

Anterior cruciate ligament (ACL) injuries are relatively common and generally occur during sports such as football and basketball practices and games [1], and symptoms such as pain and limited movement may occur after the injury. Studies have shown that ACL injuries occur approximately 17.5 times per 100,000 people/year [2] and that noncontact injuries are the main cause of ACL injuries. Noncontact injuries account for 55% of the total number of injuries in adults and 68% in adolescents, and the incidence rate in females is higher than that in males [3, 4]. ACL injury not only limits the ability to perform activities of daily living and brings about a very large economic burden caused by surgery and rehabilitation expenses but also leads to a long recovery period for returning to work and sports, resulting in a large societal burden [5, 6].

For the treatment of ACL injuries, studies have shown that anterior cruciate ligament reconstruction (ACLR) may be superior to conservative treatment for improving function, while conservative treatment is considered more appropriate for participants with neither high activity demands nor associated symptoms [7]. However, many challenges are faced in rehabilitation after ACLR. ACLR often results in changes in the movement patterns of injured and noninjured legs, such as insufficient postoperative spontaneous activation of the quadriceps muscle and subsequent neuromuscular control disorders that are common after surgery [8]. This impairment often lasts for several years and increases the risk of noncontact ACL injury in the future [9–13]. Some studies revealed that after ACLR, the risk of a second ACL injury in participants was increased by more than 20% [14, 15]. Therefore, neuromuscular control of the quadriceps is a modifiable factor, and early activation treatment is highly important for preventing second ACL injuries and promoting the participants' return to sports [13, 16].

With respect to the causes of insufficient spontaneous activation of the quadriceps muscle in the early stage after ACLR, previous mechanistic studies were based mainly on arthrogenous muscle inhibition. Based on this mechanism, previous studies have shown the efficacy of treatments such as cryotherapy, exercise training, and low-frequency electrical stimulation [17, 18]. After anterior cruciate ligament reconstruction, the excitability of the spinal reflex pathway increases, while that of the corticospinal pathway decreases. These changes coexist with the reduction of quadriceps muscle strength and voluntary activation [19–22]. Specifically, owing to the loss of mechanoreceptors in the ACL, knee joint effusion and pain and sharp changes in the excitability of afferent nerves and reflexes can be observed after injury [20]. Although ACL injury does not cause direct damage to the central nervous system, the interruption of afferent input from injured ACL mechanoreceptors and the related movement compensation results in changes in the activity and excitability of the motor cortex and corticospinal tract. The current meta-analysis indicates that the motor thresholds of both the reconstructed and non-reconstructed legs of ACL reconstruction participants are significantly increased compared with those of healthy control legs. Compared with the unreconstructed leg, the motion threshold of the reconstructed leg was also significantly increased [19]. This reduction in corticospinal excitability may reduce the neural drive of the quadriceps during maximum voluntary contraction and result in insufficient spontaneous activation of the quadriceps as well as a continuous decrease in the strength of the quadriceps after ACLR. In addition, studies have revealed that the abnormal activation and excitability of the descending corticospinal pathway are slightly different from the changes in corticospinal excitability. However, there was no significant difference in spinal reflex excitability between the healthy side and the affected side. These systemic changes in the function of the peripheral and central nervous systems may cause traditional treatment methods to have little effect on participants with these injuries, which may lead to poor clinical outcomes [23–26]. Studies have shown that anodic transcranial direct current stimulation (tDCS) effectively reduces the inhibition and promotion of quadriceps maladjustment after ACLR surgery, confirming that anodic transcranial direct current stimulation (tDCS) can improve the promotion and inhibition after ACLR, and these are precisely the driving factors of articular muscle inhibition [27]. Rehabilitation interventions focused on the normalization of the excitability of neural pathways may be an effective method of resolving quadriceps dysfunction after ACLR [28].

Transcranial magnetic stimulation (TMS) is a noninvasive brain stimulation method that can painlessly induce motor evoked potentials (MEPs). In TMS, current is discharged in an insulated coil placed on the surface of the scalp to generate a brief focused magnetic field that can depolarize neurons in the underlying cortical area through electromagnetic induction. Delivery of repetitive transcranial magnetic stimulation (rTMS) can achieve continuous regulation of neuronal excitability in the target cortical structure [29, 30]. Currently, rTMS is widely used to promote motor function recovery in participants with stroke and spinal cord injury (SCI) by regulating the excitability of the motor cortex, and rTMS has been shown in relevant studies to promote motor function, including muscle strength [31]. In addition, TMS has positive effects on the activation of the quadriceps and maximum voluntary contraction (MVC) ability [32].

It is currently unclear whether rTMS can be used to promote spontaneous activation of the quadriceps and recovery of maximum muscle strength in the early or acute stages after ACLR. Therefore, in this study rTMS was used to increase the excitability of the corticospinal pathway. The aim of this study was to explore whether rTMS could promote quadriceps activation, prevent muscle atrophy, and accelerate muscle strength recovery in participants in the early postoperative period after ACLR, thereby shortening the rehabilitation time of participants and reducing the risk of re-injury.

Methods

Research and design

This randomized controlled clinical trial was conducted in accordance with the requirements of the Comprehensive Trial Reporting Standards (CONSORT) and the principles of the Declaration of Helsinki. The study was approved by the Clinical Research Ethics Committee of Wuxi Ninth People's Hospital Affiliated with Soochow University (approval number: KS2023040, 28/08/2023). The Registration number of the Chinese Clinical Trial Registry is ChiCTR 2,300,076,522, 11/10/2023. Sixty participants who underwent arthroscopic ACLR at the Department of Orthopedics of Wuxi 9th Affiliated Hospital of Soochow University between October 7, 2023, and March 06, 2024, were enrolled as study participants and randomly assigned to the intervention group (REAL group) or the control group (SHAM group) according to a random number table. All the surgeries were performed by the same surgeon and his assistant. The surgeon has over 10 years of surgical experience, and the assistant has over 5 years of working experience. A 20 mm graft was prepared by removing the peroneus longus and semitendinosus muscles under arthroscopy. A Kirschner wire was used in conjunction with a 4.5 mm drill to penetrate the lateral cortex to create a femoral tunnel, and the graft was then fixed. The randomization order is generated by a computer, and the randomized blocks are stored in sequentially numbered sealed opaque envelopes. At the same time, the evaluators and statisticians involved in the study were kept blind.

Participants

Each patient's eligibility for inclusion was assessed by a physiotherapist. All the patients were ordinary people, not athletes, and had no regular exercise habits. The researchers provided written information about the trial protocol to participants who met the inclusion criteria and showed an interest in getting involved. Researchers are required to sign a written informed consent form.

The inclusion criteria were as follows: (1) Participants aged between 18 and 40 years; (2) participants whose grafts were from the autologous semitendinosus tendon and peroneus longus tendon; (3) The participants had a history of chronic injuries such as excessive joint movement, long-term weight-bearing or poor healing after acute injuries, resulting in damage to the articular cartilage and joint surfaces, as well as damage to the surrounding tissues of the joints (4) participants who had not received rehabilitation training before surgery; (5) participants with no cognitive abnormalities who could correctly perform home rehabilitation exercises; and (6) participants who understood all matters regarding the study before the surgery and signed the informed consent form. The following exclusion criteria were used: (1) Participants with a previous serious injury or history of surgery on the side of the affected knee joint; (2) participants with a lower extremity fracture; (3) participants with an intracranial metal or magnetic implant and a pacemaker or other medical device implants; (4) participants with a history of head injury or fracture; (5) participants with a history of epilepsy, depression, or migraine; (6) Participants with organic brain disorders such as brain tumors, recent brain hemorrhage or infarction, unstable period of brain trauma, severe headache or increased intracranial pressure; (7) Participants with hearing sensitivity or hearing impairment; (8) Participants were in the pregnancy and lactation periods; and (9) participants who could not fully extend the knee joint passively before surgery.

Measures

Routine treatment and home training guidance for all participants were provided by one physical therapist in our hospital who had been engaged in rehabilitation therapy after ACLR for more than 9 years. TMS treatment was administered by a physical therapist who has been engaged in this work for 3 years. The intervention personnel were not aware of the content and purpose of the study. The evaluation was performed by a rehabilitation physician and a physical therapist; the evaluators were blinded to the grouping and intervention methods. The schedule of rehabilitation treatment for the participants in the two groups is presented in Table 1.

Rehabilitation education

After ACLR, participants were recommended to avoid weight-bearing training on the affected leg for 2 weeks, resistance knee flexion training for 6 weeks, and the following actions on the affected leg for 3 months: rapid kicks, single-leg jumps, sudden stop-and-go motions, and rapid changes in direction. Participants were informed

Table 1 Time course of rehabilitation treatment for the participants in the two groups

	SHAM group					REAL group				
Postoperative time (days)	1	2–6	7–8	9–13	14–15	1	2–6	7–8	9–13	14–15
Postoperative rehabilitation education	√					√				
Routine muscle strength training		√	√	√	√		√	√	√	√
Passive knee extension exercise		√	√	√	√		√	√	√	√
Passive knee flexion exercise			√	√	√			√	√	√
Muscle strength training		√		√			√		√	
Repetitive transcranial magnetic stimulation							√		√	

that it is normal to have mild to moderate pain during training and that if the pain is relieved half an hour after the training, no injury has been caused. Participants were also informed that joint swelling and elevated skin temperature are normal phenomena during training and can be relieved by applying cold compresses. Finally, participants were told that if joint swelling was severe or the skin temperature remained high (2 °C higher than the healthy side), the training should be stopped, and the patient should visit the clinic in a timely manner.

All participants received guidance on how to perform self-rehabilitation training exercises, which started from the second day after surgery and lasted for 2 weeks. The training exercises were as follows: (1) Ankle pump exercise: While in the supine position, the patient held bilateral ankle plantar flexion and dorsiflexion for 4 s each. Three sets of 20 repetitions were performed 3 times a day, with a 1-minute rest period between sets. (2) Quadriceps contraction exercise: While in the supine position, the patient placed a rolled towel with a thickness of 10 cm under the popliteal fossa on both sides, and the patient applied force through the vastus medialis oblique to press the rolled towel under the popliteal fossa while lifting the distal calf. The position was maintained for 10 s each time. Three sets of 15 repetitions were performed 3 times a day, with a 1-minute rest period between sets. (3) Isometric contraction of hamstring muscles: While in the supine position, the patient tried to exert force to move their heel toward their buttocks, but the force to achieve this range of motion could not be generated. The patient tried to exert for 10 s each time, with the exerted force not causing pain in the knee joint on the affected side. Three sets of ten repetitions were performed once a day, with a 1-minute rest period between sets. (4) Straight-leg lifting exercise: While in the supine position, the patient attempted to lift the ankle joint on the affected side approximately 20 cm while extending the affected knee as much as possible. The patient was told to hold the position for 10 s each time. Three sets of ten repetitions were performed once a day, with a 1-minute rest period between sets. (5) Side-lying hip abduction: The patient was in the lateral position with the affected leg on top. Three sets of 10 repetitions of hip abduction held for 10 s were performed in three positions: a 15° hip flexion, a neutral position and a 15° hip extension. The distal lower extremity was elevated by approximately 20 cm. The rest period between sets was 1 min. (6) Passive knee joint extension exercise: The patient performed this training once a day. While in the supine position, the patient passively elevated the heel on the affected side using a pillow until the popliteal fossa was suspended 5 cm above the surface. The patient naturally relaxed the lower limb on the affected side, allowing the weight of the lower limb to stretch the knee joint to increase its extension range of motion. This position was maintained for 30 min each time.

One week after surgery, a knee flexion training exercise was added once a day. The patient was in a sitting position, with their thigh and buttocks placed on the bed and the knee joint and calf hanging off the side of the bed. Three repetitions of knee joint flexion to the 60° position (the angle of knee flexion training was gradually increased every week) was performed each time for 2 mins, with a 1-minute rest period between sets. After training, the patient was required to apply a cold compress for approximately 15 mins.

Conventional treatment

Conventional treatment started on the 2nd day after surgery and continued for 2 weeks. Under the supervision of a rehabilitation therapist, the patient underwent quadriceps strength training 5 times a week. Before performing these exercises, two crushed ice packs were fixed with an elastic bandage to the front and back of the knee, and this cold compress treatment was applied for 20 mins. Under the supervision of a therapist, the patient performed three sets of the quadriceps contraction exercise and straight-leg lifting exercise [33].

Intervention

For the REAL group, the supervised strength training exercises administered to the SHAM group were replaced with rTMS starting from postoperative day 2, and the participants were asked to perform the maximum voluntary quadriceps contraction exercise while thigh muscle twitching was visible. After rTMS was completed, three sets of the straight-leg lifting exercise were completed under the supervision of a therapist. The above interventions lasted 2 weeks.

rTMS therapy was performed as follows: Participants sat or reclined comfortably in a chair with a neck rest cushion. Stimulation was applied by aligning the center of the double-cone coil (VISHEE Magneuro100) with the primary motor cortex on the healthy side (4 ~ 6 cm opened laterally to the intersection of the line connecting the anterior dimples and the craniocerebral midline). The specific coil position for each course of treatment was determined by the significance of the tingling sensation/twitching contraction in the thigh musculature. The stimulation parameters were as follows: 45 sets of 40 pulses at 20 Hz at 100% resting motor threshold (RMT), with an interval of 28 s between sets. A total of 1800 pulses were delivered in a total stimulation time of approximately 22 min [34]. The RMT was defined as the minimum stimulator intensity required to elicit 5 MEPs with an amplitude ≥ 50 μV in 10 trials.The MEP data were recorded using a surface electromyography system. The surface electrodes were placed on the belly of the tibial anterior muscle on the affected side.

Adherence monitoring program

During the initial diagnosis and treatment, the patient received a prescribed exercise plan form for home training. The form included the training method; the time, frequency, and intensity of exercises; precautions; and a signature and check-off section to record the completion of the training. If less than 75% of the prescribed training plan was completed, the patient was excluded from the statistical analysis.

In this study, follow-up evaluations of participants at 2 days and 15 days after surgery were conducted, and participants lost to follow-up were also excluded from the statistical analysis.

Outcome indicators

Primary outcome indicators

(1) RMT: The MEP evoked by the TMS device at 2 and 15 days after surgery and the changes in motor cortex excitability and corticospinal and intracortical excitability were evaluated. The subject sat in a comfortable chair, with their body relaxed and their hands placed at their sides. TMS was administered through a double-cone coil. The coil was tangent to the corresponding motor cortex of the lower extremity at the most appropriate stimulation point. The RMT was considered the minimum strength that evoked an MEP with a peak amplitude greater than 50 μV in the resting target muscle on no less than 5 out of 10 trials. If an MEP was not induced, the RMT was recorded as zero. The RMT value of the vastus medialis oblique on the affected side was measured and recorded.

(2) Root-mean-square (RMS) value: The RMS value can represent the average intensity of the electromyographic (EMG) signal, and it is the most reliable parameter in the EMG time domain. The RMS value can be used to evaluate muscle strength during exercise and in the resting state. The examiner evaluated the RMS of the patient on the 2nd and 15th days after surgery. In this study, a two-channel electromyographic (EMG) feedback therapy device (Model S4 30, VISHEE Company, China) was used to analyze the collected raw EMG signals with its supporting surface EMG test function. The inner thighs of the participants were wiped with 75% ethanol. A pair of test electrodes was placed at the fullest part of the vastus medialis oblique, with the electrodes aligned with the direction of the muscle fibers of this muscle. The distance between the electrodes was > 1 cm. The reference electrode was placed at the protrusion of the bony landmark lateral to the test electrode. The ambient temperature was maintained at 26–28 °C. The patient remained awake and relaxed in the supine position, with the femur in a neutral rotation position. The examiner instructed the patient to try to lift their calf upward to the maximum extent while actively performing subpopliteal compression so that their heel could be lifted off the bed surface. The RMS on the affected side was measured. We record the RMS values three times and then take the average.

Secondary outcome indicators

(1) Difference in the circumference of the vastus medialis oblique: At 2 and 15 days after surgery, the examiner measured the circumference of the vastus medialis oblique with a circumference ruler and asked the patient to actively contract the vastus medialis oblique on the healthy side to locate the highest point of muscle belly prominence. The distance from this point to the medial femoral condyle was measured, and then the difference in circumference between the two sides at this position was measured in the relaxed state (circumference on the healthy side minus the circumference on the affected side) to determine the muscle atrophy under different experimental conditions.

(2) Differences in the extension angle between the healthy and affected sides: At 2 and 15 days after surgery, the examiners used a range-of-motion measuring ruler to measure the active extension range of the knee joint on the healthy side and the affected knee joint, respectively, with the patient in

a sitting position and the knee joint at 60° of flexion. The measuring axis was the lateral femoral condyle, the fixed arm was parallel to the long axis of the femur, and the mobile arm was parallel to the long axis of the calf. The patient was asked to actively try to extend their knee joint to the maximum hyperextension position. The examiner recorded the angle. The measured angle was taken as a positive value when hyperextension was reached and a negative value when the knee was underextended. The difference in knee extension angles between the healthy and affected sides was obtained by subtracting the extension angle of the healthy knee joint from the extension angle of the affected knee joint.

(3) Visual analog scale (VAS): Participants used the VAS to rate their average pain intensity on the past day. We used a VAS of 0 mm(no pain) to 100 mm(maximum pain). VAS scores were evaluated 2 days and 15 days after surgery.

Statistical analysis

The sample size was calculated using PASS23 software based on the RMT results of a pre-experiment study. Each group requires 23 participants to achieve a 90% test power at a P value of 0.05. Considering the expected dropout rate of 20%, the final number of participants was determined to be 60 [35].

The statistical analysis was performed by a statistician who was completely independent of this study, and SPSS 26.0 software was used for the statistical analysis. Continuous variables are presented as the mean ± standard deviation (SD), and categorical variables are presented as frequencies or percentages. The comparison of continuous variables of the basic information of the two groups of participants was performed via an independent samples t test, and the comparison of categorical variables was performed via the chi-square test. Comparisons of the RMT, RMS value, difference in the circumference of the vastus medialis oblique, and difference in the active extension between the healthy and affected sides between the two groups were performed via paired t tests, while intergroup comparisons of these values before and after treatment were performed using independent t tests. The statistical significance threshold was set to 0.05 (two-sided).

Results

Participants

Sixty participants were randomly assigned to the REAL group or the SHAM group. 3 participants in the SHAM group and 1 patient in the REAL group were excluded from the trial because of insufficient check-in and loss to follow-up. A total of 56 participants successfully completed the baseline assessment, the 2-week intervention and the post-treatment assessment, and no adverse events were observed during the intervention (Fig. 1). Before the intervention, there were no significant differences in demographic or clinical characteristics between the two groups (Table 2).

Primary outcomes

After two weeks of treatment intervention, within-group analyses showed that RMT and RMS values significantly improved in both groups from baseline to post-treatment (all $P < 0.05$; for RMT in the SHAM group, $P = 0.027$; all other P values < 0.001). There was a statistically significant difference between the two groups after treatment ($P < 0.001$) (Table 3).

Secondary outcomes

At baseline and after intervention, the changes in circumference difference, knee extension lag angle, and VAS in both groups were statistically significant (all $P < 0.05$, except for VAS in the SHAM group, $P = 0.009$, and the rest $P < 0.001$). In the assessment at the end of treatment, statistical analysis showed that there were statistically significant differences between the two groups in knee extension lag angle and VAS ($P < 0.001$), but no significant statistical difference in the circumference difference of the healthy and affected sides (Table 3).

This indicates that rTMS is significantly superior to the SHAM intervention in improving quadriceps function recovery of quadriceps function in participants after anterior cruciate ligament surgery, but there is no statistical significance in terms of circumference (Table 3).

Discussion

This study explores the effect of transcranial magnetic stimulation on the quadriceps femoris from a central nervous system perspective. It may provide new diagnostic and therapeutic ideas for subsequent clinical rehabilitation treatments. The study results indicate that rTMS can be used as an adjuvant therapy in primary rehabilitation in the early postoperative period after ACLR and can improve clinically evaluated quadriceps muscle function and the RMT. These findings may be of great significance, as insufficient activity of the quadriceps muscles is a major problem faced by patients after ACLR surgery, and the current treatment strategies have varying levels of effectiveness.

However, a clear and clinically important difference in quadriceps muscle strength recovery in the long term (more than 2 weeks) was not identified in the present study. Compared with conventional strength training, the added rTMS treatment was significantly associated with a large increase in the surface EMG signals of the quadriceps femoris, and a significant decrease in the

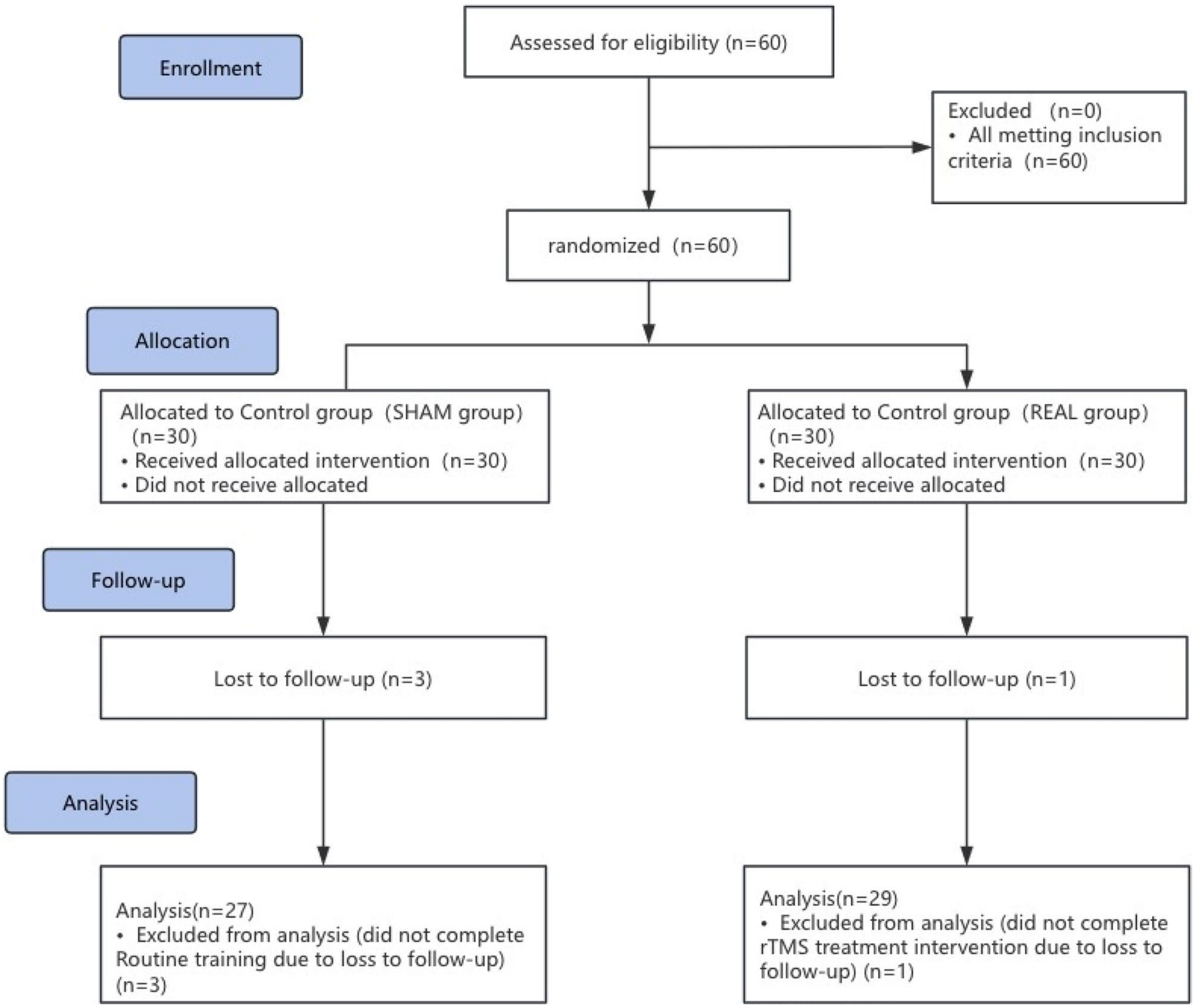

Fig. 1 Flowchart of included participants

Table 2 General clinical data of the participants in the two groups

Group	Number of participants	Sex (participants)		Average age (years, $\bar{x}$ ±s)	Surgical side (participants)		Mean body mass index (kg/m^2, $\bar{x}$±s)
		Male	Female		Left side	Right side	
SHAM group	27	20	7	28.96 ± 8.89	16	11	24.48 ± 3.16
REAL group	29	22	7	29.31 ± 6.89	15	14	24.16 ± 4.27

RMT was observed. The changes in these two primary outcome indicators indicate that rTMS treatment greatly improved the muscle function of the quadriceps femoris and that during exercise and in the resting state [22]. rTMS is also conducive to long-term improvements in quadriceps function. However, there was no significant difference in the circumference difference of the medial head muscle of the quadriceps between the two groups, which may be due to the better postoperative swelling reduction effect and lower muscle atrophy in the REAL group, which combined to cause little difference in circumference compared with the SHAM group. Given that edema has multiple causes and is difficult to control, future studies could compare the cross-sectional muscle volume of the vastus medialis oblique between the two groups using magnetic resonance imaging (MRI) to better exclude the confounding factor of preoperative and postoperative edema [36]. Then, the effects of early rTMS treatment on quadriceps muscle mass after ACLR could be explored more intuitively.

For participants who undergo ACLR, activation of the quadriceps muscle, especially the vastus medialis oblique, has always been a challenge for clinical medical personnel in the early postoperative period. In the past,

Table 3 Comparison of the RMT, RMS, circumference difference, knee extension lag angles, VAS in the two groups before and after treatment

Outcome	Time	Group	Mean ± SD	Mean Difference (SHAM - REAL)	95% CI for Difference	t/z	df	*p*	Effect Size
RMT	Before	SHAM	51.96 ± 15.05	6.24	(−0.98, 13.46)	1.738	48.44	0.088	d = 0.47
		REAL	45.72 ± 11.43						
	After	SHAM	43.41 ± 16.62	14.00	(6.30, 21.70)	3.666	45.25	< 0.001	d = 0.99
		REAL	29.41 ± 11.24						
Circumference difference	Before	SHAM	−1.307 ± 1.39	−0.047	(−0.71, 0.62)	−0.142	48.28	0.889	d = −0.04
		REAL	−1.26 ± 1.05						
	After	SHAM	−0.200 ± 1.57	−0.59	(−1.29, 0.11)	−1.708	40.80	0.095	d = −0.47
		REAL	0.39 ± 0.90						
Knee extension lag angles	Before	SHAM	11.02 ± 4.43	1.48	(−1.00, 3.96)	1.192	54.00	0.238	d = 0.32
		REAL	9.54 ± 4.86						
	After	SHAM	8.46 ± 3.69	4.97	(3.27, 6.67)	5.885	44.80	< 0.001	d = 1.60
		REAL	3.49 ± 2.46						
RMS	Before	SHAM	11.47(7.62, 14.28)	−0.74	(−3.72, 2.48)	0.42	41	0.674	*r* = −0.06
		REAL	12.21 (8.18, 16.99)						
	After	SHAM	18.73(13.33, 23.86)	19.22	(−25.81, −13.15)	4.91	41	< 0.001	*r* = −0.66
		REAL	37.95 (28.57, 54.87)						
VAS	Before	SHAM	3 (2, 3)	1	(0.0, 1.0)	1.47	41	0.141	*r* = −0.20
		REAL	2 (2, 3)						
	After	SHAM	2 (1, 3)	1	(1.0, 1.0)	4.92	41	< 0.001	*r* = −0.66
		REAL	1 (0, 1)						

Independent t-tests were used for between-group (SHAM vs. REAL) comparisons at each time point. Mean difference = SHAM minus REAL. Positive values indicate higher values in SHAM group. 95% CI and p reflect Welch's correction for unequal variances where applicable

many related studies, such as those on cryotherapy and other means of therapy, have been conducted. Inhibition of the intra-articular inflammatory response to improve the recruitment of muscle fibers, or direct stimulation of muscle fibers through low-frequency electrical stimulation and electromyography feedback, is considered to improve the activation performance of this muscle to some extent [37]. With respect to studies of the excitability of the cerebral cortex and corticospinal pathway after reconstruction [38–41], most of these studies were limited to the knee joint or the quadriceps muscle itself and lacked evaluation of brain regulation. As a noninvasive brain stimulation method, TMS has been widely used in the fields of brain injury and SCI. Many studies have shown that the effects of TMS on the motor function of the lower extremities of participants with brain and spinal cord injuries include therapeutic effects on quadriceps strength and other aspects [42]. Compared with that of brain and SCI, the conduction pathway of ACLR is not damaged, and the only manifestation is excitability changes. TMS has previously been used to stimulate the relevant brain regions of participants undergoing ACLR in only a few previous studies. Although studies have shown that brain stimulation can significantly enhance the excitability of the corticospinal tract and improve the function of the quadriceps muscles, there are relatively few studies on stimulating the relevant brain regions in the early postoperative period on the second day [27]. Therefore, we attempted to enhance the excitability of the corticospinal pathway and the strength of the quadriceps muscle by using repetitive transcranial magnetic stimulation during the acute phase after ACLR.

In 2019, C. Krishnan reported the case of a patient who underwent muscle contraction training combined with TMS in the 6 months after ACLR, and the corticospinal pathway excitability of the quadriceps muscle could be increased during the training process while increasing quadriceps strength, spontaneous activation, and limb symmetry. In this study, rTMS treatment was performed in addition to strength training in the acute stage (the second day after surgery). Although the intervention lasted only two weeks, there was significant improvement in quadriceps muscle strength and other aspects. In recent years, the population of participants with ACL injuries has become increasingly younger [1], and the demand for participants to return to sports after ACLR has also increased. We will conduct a study with a longer follow-up to observe the muscle strength performance of participants at 3, 6 and even 9 months. The criteria for considering a return to sports should include that the maximal muscle strength of the quadriceps and hamstrings in the participants reaches more than 90% of that of the healthy side. Through longer follow-up studies, it can be determined whether this type of intervention can shorten the time to return to sports and the course of rehabilitation. For participants in the early postoperative period after ACLR, early intervention to promote the excitability of the cerebral cortex and corticospinal pathway may have long-term effects on quadriceps strength.

In addition, even if the rTMS effect is better than that of conventional strength training, the intervention duration used in this study may not be the most appropriate choice. Some studies have shown that 4 weeks of adjuvant rTMS therapy during primary rehabilitation after SCI can promote long-term recovery of lower extremity muscle function [43]. In future studies, different rTMS intervention durations should be set as a grouping variable to explore the most appropriate rTMS intervention duration for participants in the early postoperative period after ACLR. In addition, the current studies of rTMS intervention programs lack explorations of TMS parameters. Parameters such as the frequency and pulse of rTMS intervention also affect muscle strength [29]. Given the importance of these parameters, the parameters of rTMS should also be evaluated accordingly in future studies to observe the differences in the effects of different parameters on the quadriceps muscle fiber recruitment effect.

This study evaluated the combined effects of direct strength training and rehabilitation exercises and rTMS treatment, which specifically targets corticospinal pathway excitability, and the study revealed that rTMS treatment was able to increase patient corticospinal pathway excitability, which was associated with increases in spontaneous activation and quadriceps strength, both of which are very beneficial for participants. However, rTMS treatment is unlikely to replace direct high-intensity quadriceps strengthening techniques (such as eccentric contraction or neuromuscular electrical stimulation). In contrast, we believe that this training can be used together with strength training as adjuvant therapy. The advantage of rTMS is that it can be used in the acute stage of recovery after ACLR, avoiding injury to the graft or patella fracture that may be caused by traditional high-intensity intervention measures. Furthermore, this study is one of the first to conduct repetitive transcranial magnetic stimulation (rTMS) on the corticospinal pathway to address the early-stage quadriceps muscle inhibition problem after ACLR. This period is a crucial stage for preventing muscle atrophy and functional impairment. This study provides evidence that rTMS can be a valuable adjunctive tool in the immediate postoperative phase after ACLR. By rapidly enhancing quadriceps activation and reducing extension lag, it can help athletes initiate and progress through foundational strength training sooner. By overcoming neural inhibition, rTMS may allow athletes to engage more effectively with voluntary exercises prescribed during early rehabilitation, potentially maximizing the benefits of standard physiotherapy protocols. By addressing fundamental neural and functional deficits early (improved activation, reduced extension lag, less pain), rTMS may contribute to shortening the overall timeline for achieving key milestones in the return-to-sport continuum, although long-term studies are needed to confirm this.

Study limitations

This study has the following limitations: (1) This study is a single-center trial. Currently, the sample size is insufficient to conduct subgroup analyzes based on covariates such as disease duration and severity of injury; (2) We collected data to evaluate the short-term benefits of the proposed intervention measures but did not assess the long-term effects of the two-week training on the quadriceps muscle function of participants with anterior cruciate ligament reconstruction.(3) This study did not investigate the parameters of repetitive transcranial magnetic stimulation (rTMS), nor did it set different durations of rTMS intervention as a grouping variable to explore the most appropriate rTMS intervention duration for participants in the early stage after ACLR. In the future, we will conduct follow-up visits for these participants and explore the long-term effects of this study on the quadriceps muscle function. In conclusion, it is necessary to conduct further large-scale clinical trials to verify the effectiveness of this comprehensive treatment method. Additionally, we will continue to expand the sample size and conduct long-term follow-up evaluations to promote the development of this new rehabilitation treatment technology.

Conclusions

The research results indicate that repetitive transcranial magnetic stimulation (rTMS) aimed at enhancing the excitability of the corticospinal tract may be able to promote the improvement of quadriceps muscle function in early post-ACLR participants, prevent muscle atrophy, and accelerate the recovery of muscle strength.In future studies on the effect of rTMS on the recovery of maximum muscle strength in participants early after ACLR, a longer intervention period should be used, a larger sample should be evaluated, and the potential effects on training capacity and intensity should be studied.

Abbreviations

ACL	Anterior cruciate ligament
ACLR	Anterior cruciate ligament reconstruction
EMG	Electromyographic
MEPs	Motor evoked potentials
MVC	Maximum voluntary contraction
rTMS	Repetitive transcranial magnetic stimulation
RMS	Root mean square

RMT Resting motor threshold
SCI Spinal cord injury
TMS Transcranial magnetic stimulation
VAS Visual analog scale

Supplementary Information

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Supplementary Material 1

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Not applicable.

Authors' contributions
All authors contributed to the study. Simiao Zhang: Methodology, Writing – original draft. Ying Zhao: Writing – review & editing. Wenwen Li: Data curation & editing. Ruijun Bai: Data curation & review. Chengwei Shi: Data collection& editing. Jilong Han: review & editing. Yu Zhang: review & editing. *Jingyi Mi: Data curation & review.

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Data availability
The datasets used and/or analyzed during the current study are available from the first or corresponding author on reasonable request.

Declarations

Ethical approval and consent to participate
The study was approved by the Clinical Research Ethics Committee of Wuxi Ninth People's Hospital Affiliated with Soochow University (approval number: KS2023040, 28/08/2023). The Registration number of the Chinese Clinical Trial Registry is ChiCTR2300076522, 11/10/2023. All methods were performed according to the "Declaration of Helsinki" by the ethical guideline and regulations.
All the participants agreed to participate in this study and signed the informed consent form.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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